Ameer Ogirimah

Senior AI & Software Engineer

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PROFESSIONAL SUMMARY

Senior AI & Software Engineer with 10+ years of experience architecting high-impact, production-grade AI platforms, real-time media streaming backends, and distributed enterprise automation systems. Deep expertise in LLM architectures, agentic AI workflows, RLHF post-training, MLOps, and RAG pipelines, backed by full-stack software engineering mastery across Python, TypeScript/React, JavaScript, Rust, and SQL. Proven track record slashing financial decision turnaround by 97% and clinical onboarding overhead by 65% at Xylont, evaluating and post-training frontier LLMs at Outlier AI, and engineering automated Python/FFmpeg availability monitoring and idempotent scheduling engines at Steam and Globe and Gotel Communication.

CORE TECHNICAL SKILLS

AI & Machine Learning: LLM Fine-Tuning & RLHF Evaluation, RAG Architectures, Agentic AI, Multi-Agent Orchestration, Prompt & Context Engineering, LangChain, LangGraph, LlamaIndex, PyTorch, Speculative Decoding

Languages & Frameworks: Python (FastAPI, Flask, Django, Celery, PyTorch, asyncio), TypeScript & JavaScript (React, Node.js), SQL, Rust

Cloud, DevOps & Infrastructure: AWS, GCP, Docker, Kubernetes, GitHub Actions, CI/CD, GPU Infrastructure, Linux/Unix Administration

Databases, Caching & Messaging: PostgreSQL (pgvector), SQL Server, Redis, RabbitMQ, Vector Databases

Streaming, Telemetry & Systems: FFmpeg, GStreamer (PyGObject), Real-Time Transport Protocol (RTP), WebSockets, RESTful APIs, gRPC, Prometheus, Grafana, Idempotent System Design

Engineering Practices: Microservices, Event-Driven Architecture, MLOps, Test-Driven Development (TDD), Agile/Scrum, Code Reviews, Responsible AI, Model Governance

PROFESSIONAL EXPERIENCE

Technical Manager | Xylont

Jan 2026 – Present

- Spearheaded technical architecture and end-to-end software delivery at the intersection of artificial intelligence, backend engineering, and business operations, deploying production-grade AI platforms that increased operational throughput by 3.5x.
- Architected and deployed an automated financial record evaluation and decision-support engine using multi-agent LLM workflows (LangGraph, FastAPI) and asynchronous document parsing, reducing average end-to-end processing time from 4 days to 120 minutes (a 97% reduction) and cutting manual review costs by 70%.
- Engineered an event-driven patient onboarding, automated triage, and multi-channel follow-up software platform for a home healthcare clinic using Python, React, and PostgreSQL, reducing administrative onboarding time by 65% and improving patient follow-up adherence by 48%.
- Designed custom MCP servers and a scalable Agentic RAG context-management layer leveraging pgvector and semantic caching, lowering LLM token consumption by 50%, reducing P95 query latency by 42%, and cutting hallucination rates across clinical and financial queries to under 2%.
- Implemented an automated MLOps observability and evaluation pipeline with structured reasoning trace logging and CI/CD guardrails, shrinking model deployment and validation cycles from days to under 2 hours (85% faster release velocity).

AI Specialist | Outlier AI

Jan 2024 – May 2025

- Audited, benchmarked, and refactored over 1,200 complex AI-generated code implementations across Python, JavaScript, TypeScript, SQL, and C, resolving algorithmic bottlenecks and edge-case bugs to boost model pass@1 code-synthesis accuracy by 34%.
- Evaluated and optimised the programmatic integration of frontier LLMs with external developer APIs, CLI utilities, and sandboxed code execution environments, improving autonomous agent tool-calling reliability by 40% and reducing redundant execution steps by 28%.
- Engineered automated Python validation harnesses, golden reference codebases, and multi-dimensional evaluation rubrics for RLHF post-training across software engineering and computational science domains, accelerating dataset QA throughput by 45%.
- Conducted deep-dive failure-mode analysis on multi-step chain-of-thought reasoning traces, identifying systemic logic and syntax regressions that improved post-trained model adherence to complex system prompts by 30%.

Information Technology Officer | University of Hertfordshire Sep 2023 – Feb 2024

- Engineered Python and Bash diagnostic automation scripts to streamline first-line software, network, and identity-management troubleshooting across academic departments, reducing average ticket resolution time by 38%.
- Nominated for the prestigious Vice-Chancellor's Team of the Year Award in recognition of outstanding technical service delivery, cross-departmental systems support, and measurable contributions to university-wide operational efficiency.
- Resolved 95%+ of complex software configuration, Virtual Learning Environment (VLE) integration, and campus authentication incidents on first contact, earning formal commendations from faculty and students for technical precision.
- Consistently exceeded strict Service Level Agreements (SLAs) by 25% across high-volume support queues, authoring standardised troubleshooting runbooks that reduced recurring software support escalations by 30%.

Senior Production Engineer | Steam and Globe Communication Dec 2017 – Sep 2022

- Architected and managed an automated 24/7 system availability monitoring, anomaly detection, and reporting platform using Python (asyncio, Celery, FastAPI/Flask), FFmpeg, and PostgreSQL to continuously probe live audio/video streams for packet loss, silence, and frame freezes—reducing unplanned system downtime by 65% and cutting Mean Time to Detect (MTTD) from 25 minutes to under 15 seconds.
- Built a real-time operational telemetry and automated incident-reporting web dashboard using Python, Redis, and WebSockets, automating daily SLA compliance reporting and reducing manual diagnostic overhead by 70%.
- Engineered low-latency Real-Time Transport Protocol (RTP) media streaming pipelines across three production channels using Python bindings for GStreamer and FFmpeg transcoding profiles, improving stream stability by 55% and reducing CPU encoding overhead by 35%.
- Developed a fault-tolerant Python microservice and TCP/serial socket daemon to automate real-time metadata ingestion and dynamic data synchronisation across three platform endpoints, increasing end-user content engagement by 42%.
- Mentored, onboarded, and trained 10+ junior software and systems engineers in Python automation, Linux server administration, Git workflows, and automated unit testing (pytest), reducing new-hire ramp-up time by 50% and improving team deployment velocity by 40%.

Systems Engineer | Gotel Communication May 2015 – Apr 2016

- Designed and engineered a mission-critical idempotent schedule ingestion component for the core programme scheduling system using Python (Django/Flask, Celery) and JavaScript, leveraging SHA-256 payload hashing, deterministic UUIDs, and Redis distributed locks with PostgreSQL transactional upserts to eliminate 100% of duplicate schedule entries during network retries and reduce scheduling conflict errors by 85%.
- Built an asynchronous client-side validation and conflict-detection interface in JavaScript integrated with RESTful Python endpoints, providing real-time timeslot collision warnings that reduced manual data-entry time by 60% and cut schedule publishing latency by 45%.
- Developed automated XML/JSON playlist parsers and background reconciler daemons in Python to synchronise schedule updates with downstream playout databases, improving end-to-end data ingestion throughput by 50%.
- Implemented automated unit and integration test suites (Python unittest) and structured application logging across the scheduling pipeline, decreasing post-release regression defects by 40% and accelerating incident root-cause analysis by 55%.

EDUCATION & CERTIFICATIONS

MSc Artificial Intelligence and Robotics (Distinction) — University of Hertfordshire, Hatfield Oct 2024

Business for All (Santander Open Academy) — Harvard Business Publishing Nov 2024

Skills for the Green Transition — Cambridge Judge Business School Dec 2023

BEng Telecommunication Engineering (Second Class Upper, 4.08/5.0) — Federal University of Technology, Minna 2014

PROFESSIONAL AFFILIATIONS & LEADERSHIP

- **British Computer Society (BCS):** Member & Membership Secretary, BCS Hertfordshire Branch (Jan 2023 – Present)
- **Institute of Electrical and Electronics Engineers (IEEE):** Member, IEEEXtreme 17.0 Ambassador; Member of the IEEE Computer Society, IEEE Computational Intelligence Society, and IEEE Robotics and Automation Society (Dec 2021 – Present)